

**ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)**  
**ORGANISATION OF ISLAMIC COOPERATION (OIC)**  
**Department of Computer Science and Engineering (CSE)**

**QUIZ 1 SET A**

**SUMMER SEMESTER, 2024-2025**

**DURATION: 20 MINUTES**

**FULL MARKS: 15**

**SWE 4805: Software Verification and Validation**

**Write all the answers on the question paper.**

Answer all **3 (three)** questions. Figures in the right margin indicate full marks of questions.

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ID:

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1.

5  
(CO1)  
(PO1)

Assume a transportation system where two types of transportation are available: bus and train. A transportation cannot be a bus and a train at the same time. Every transportation has a route, capacity, a set of platforms, next arrival, last departure, and expected travel time. This transportation system has been implemented in different stations, where every station has a name and a set of platforms. Similarly, every platform has a number, a set of transportation (bus or train), and a set of security personnel.

- Identify all possible signatures and fields in the system requirements.

#### Solution:

```

1  abstract sig Transportation {
2      route: one Route,
3      capacity: one Int,
4      nextArrival: one Time,
5      platform: some Platform,
6      lastDeparture: one Time,
7      expectedTravelTime: one Time
8  }
9
10 sig Bus extends Transportation {}
11
12 sig Train extends Transportation {}
13
14 sig Station {
15     platforms: set Platform
16 }
17
18 sig Platform {
19     number: one Int,
20     transportation: some Transportation,
21     securityPersonnel: some Person
22 }
23
24 sig Time {}
25
26 sig Person {}
27
28 sig Route {
29     assignedTransportation: one Transportation
30 }

```

2.

5  
(CO2)  
(PO2)

Consider the previously defined transportation system model. Write the following constraints in Alloy as `fact` blocks.

- (a) Transportation capacities and platform numbers must be positive.

(b) Transportation and route relationships, and platform and transportation relationships must remain consistent.

(c) Bus and train transportation systems must operate on separate platforms.

(d) Platform numbers must be unique.

(e) Transportation-platform relationships must remain consistent from both directions.

#### Solution:

```
1 fact nonZero {
2     all t: Transportation | t.capacity > 0
3     all p: Platform | p.number > 0
4 }
5 fact differentRoute {
6     all t: Transportation |
7         t in t.route.assignedTransportation
8
9     all p: Platform |
10        p in p.transportation.platform
11 }
12
13
14 fact differentPlatform {
15     all b: Bus, t: Train | no b.platform & t.platform
16 }
17 fact differentNumber {
18     all disj p, q: Platform |
19         p.number != q.number
20 }
```

```

21
22 fact transportationConsistency {
23     all t: Transportation, p: Platform |
24         p in t.platform iff t in p.transportation
25 }
26
27 run {} for 1 Station, 4 Platform, 4 Transportation, 2 Time, 2 Person, 4
    Route

```

3.

Verify and validate the following statements of the transportation system in Alloy using `assert` statements.

5  
(CO2)  
(PO2)

- (a) Every transportation must be associated with at least one platform.
- (b) No platform can simultaneously serve both bus and train transportation systems.
- (c) Different platforms must have different platform numbers.
- (d) Every transportation must have a valid route.
- (e) Transportation-platform relationships must remain consistent from both directions.
- (f) Every station must contain at least one platform.

#### Solution:

```

1
2
3
4 assert transportationPlatform {
5     all t: Transportation |
6         some t.platform
7 }
8

```

```

9
10
11 assert separatePlatform {
12     all p: Platform |
13         no ((p.transportation & Bus)
14             & (p.transportation & Train))
15 }
16
17
18
19 assert uniquePlatformNumber {
20     all disj p, q: Platform |
21         p.number != q.number
22 }
23
24
25
26 assert validRoute {
27     all t: Transportation |
28         some t.route
29 }
30
31
32
33 assert transportationConsistency {
34     all t: Transportation, p: Platform |
35         p in t.platform iff
36         t in p.transportation
37 }
38
39
40
41 assert stationPlatform {
42     all s: Station |
43         some s.platforms
44 }
45
46 check transportationPlatform for 5
47 check separatePlatform for 5
48 check uniquePlatformNumber for 5
49 check validRoute for 5
50 check transportationConsistency for 5
51 check stationPlatform for 5

```

### MARKS SUMMARY TABLE

| Question     | Marks | Keywords                    |
|--------------|-------|-----------------------------|
| 1            | 5     | Signature and Fields        |
| 2            | 5     | Facts and Constraints       |
| 3            | 5     | Assertions and Verification |
| <b>Total</b> | 15    |                             |